Status	Finished
Started	Tuesday, 20 October 2026, 10:05 PM
Completed	Tuesday, 20 October 2026, 10:10 PM
Duration	4 mins 54 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search is more efficient than binary search on large lists.
d. Linear search requires the list to be sorted.

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It compares elements sequentially
c. It always starts from the beginning of the list
d. It can be applied only if the list is sorted ✓

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By starting from the first element
c. By comparing each element sequentially
d. By using a hash function

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Bubble search
- c. Linear search ✓
- d. Quick search

Question **5**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. log n
- b. n-1
- c. n/2
- d. n ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- b. Must use a sorted array
- c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- d. There must be a mechanism to access middle element directly

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. An error is raised
- b. The first element is returned
- c. The last element is returned
- d. The search is considered unsuccessful ✓

Question **8**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear

b. Both (1) & (3)

c. Hash

d. Binary ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓

b. O(log n)

c. O(1)

d. O(n log n)

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops

b. The search continues in the left or right sublist ✓

c. The list is sorted

d. The search continues from the beginning

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list

b. In the right sublist

c. In the entire list

d. In the left sublist ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Inserting elements into a list
c. Deleting elements from a list
d. Determining whether an element is present in a list ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Mining
c. Searching ✓
d. Discovering

Question **14**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is not the array at all
c. When the item is the last element in the array
d. When the item is somewhere in the middle of the array ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element ✓
b. The lower middle element is chosen as the middle element
c. The middle element is chosen randomly
d. The search stops